

Project Design Phase

Date	7 July 2026
Team ID	SWTID-2026-2886
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	2 Marks

Template:

Problem-Solution fit canvas 2.0

Purpose / Vision: To enable data-driven insights by analyzing and visualizing electricity consumption patterns across Indian states and regions.

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) Who is your customer? I Government energy planners - Government energy planners - State electricity boards and regulators - Power distribution companies - Energy analysts and data researchers - Policy makers and decision makers - General public and students interested in energy data	6. CUSTOMER CONSTRAINTS What constraints prevent your customers from taking action or limit their choices of solutions? i.e. spending power, budget, no cash, network connection, available devices. - Large and complex datasets are difficult to analyze manually - Lack of interactive visualization tools - Limited access to real-time electricity data insights - Time-consuming process to compare multiple years and regions - Technical expertise required to generate meaningful reports - Budget constraints for advanced analytics tools - Dependence on static reports and spreadsheets	5. AVAILABLE SOLUTIONS Which solutions are available to the customers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? i.e. pen and paper is an alternative to digital notetaking - Manual analysis using Excel spreadsheets - Traditional reporting from electricity boards - Static PDF or printed reports - Third-party BI tools (Power BI, QlikView, etc.) - Open data portals with raw datasets - Academic research papers and studies	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one; explore different sides. - Analyze electricity consumption across states, regions, and years - Compare consumption before and during COVID-19 lockdown - Identify top and bottom consuming states - Understand trends and patterns in electricity demand - Generate interactive dashboards and reports for decision making	9. PROBLEM ROOT CAUSE What is the real reason that this problem exists? What is the back story behind the need to do this job? i.e. customers have to do it because of the change in regulations. - Data is scattered across multiple sources and formats - Lack of interactive dashboards for analysis - Manual methods are slow and error-prone - No easy way to compare multiple years and regions - Limited insights about the impact of lockdown on consumption	7. BEHAVIOUR What does your customer do to address the problem and get the job done? i.e. directly related: find the right solar panel installer, calculate usage and benefits; indirectly associated: customers spend free time on volunteering work (i.e. Greenpeace) - Download datasets from government portals - Work on Excel sheets for cleaning and analysis - Create charts manually - Compare data year by year - Share reports via email or presentations - Search online for energy statistics and studies	
Identify strong TR & EM	3. TRIGGERS What triggers customers to act? i.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news. - Rising electricity demand and consumption - Need for energy planning and forecasting - Government initiatives for energy efficiency - Impact of COVID-19 lockdown on power usage - Availability of open electricity data - Academic and research requirements	10. YOUR SOLUTION If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality. If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour. - Interactive Tableau dashboard for electricity consumption analysis - Filters for State, Region, Year and Date - Top N / Bottom N and lockdown comparison - Maps, charts and KPIs for quick insights - Tableau Story for step-by-step analysis - Published on Tableau Public for sharing - Integrated with Flask web application for easy access	8. CHANNELS of BEHAVIOUR 8.1 ONLINE What kind of actions do customers take online? Extract online channels from #7 - Government open data portals - Tableau Public dashboard link - Project GitHub repository - Web application (Flask) for dashboard access - Social media and academic forums 8.2 OFFLINE What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development. - Conferences, seminars and workshops - Printed reports and presentations - Academic publications and journals	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER How do customers feel when they face a problem or a job and afterwards? i.e. lost, insecure > confident, in control - use it in your communication strategy & design. Before: Confused, overwhelmed, time-consuming, uncertain about accuracy. After: Confident, informed, satisfied, empowered to make better decisions.			

References:

- <https://www.ideahackers.network/problem-solution-fit-canvas/>
- <https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>